MODULE 5 · PREVENTION

Lesson 18: Unreliable Airspeed and Instrument Failures

Lesson 18 of 24 · about 8 min

LEARNING OBJECTIVES

- Recognise the symptoms of unreliable airspeed.
- Apply the generic memory actions: pitch and thrust.
- Describe how to troubleshoot and stabilise after unreliable air data.



Ice crystals can block pitot probes and make airspeed unreliable.

Causes

- Pitot probes blocked by ice crystals, heavy rain, volcanic ash, insects or covers left in place.
- Static ports blocked — for example by tape or residue after maintenance or washing.
- Failed probe heaters, damaged probes, or faulty air data computers.

Symptoms

- Disagreement between the captain's, first officer's and standby airspeed or altitude.
- Speed that does not match the pitch and thrust (for example, speed rising in a climb at constant thrust).
- Simultaneous, contradictory warnings: overspeed and stall warning together, or "rudder ratio", "Mach trim" and altitude alerts.
- Autopilot and autothrust disconnect, flight control law degradation on fly-by-wire types.

Memory actions

Most manufacturers' unreliable airspeed procedures begin with memory items along these lines:

1. **Autopilot and autothrust — disconnect** (they may be using the faulty data).
2. **Flight directors — off** where the procedure requires it.
3. **Pitch and thrust — set** to the memory values for the flight phase (for example, a climb attitude and climb thrust after take-off; a level-flight attitude at cruise).

Use the **exact values and procedure** from your QRH. Do not chase the airspeed indication.

Stabilise, then troubleshoot

- When safe — above minimum safe altitude, wings level, stable pitch and thrust — use the QRH tables to set pitch and thrust for the phase of flight and weight.
- Use other information: GPS ground speed, angle of attack indication where fitted, the flight path vector, and ATC for ground speed and altitude (in an emergency).
- Identify a reliable source if possible. If none, continue with pitch and power tables.

KEY LESSON

In several accidents, a short air data problem became a fatal stall because the crew reacted to the false indications with large, sustained pitch inputs. **Pitch and power first**, avoid large inputs, and respect the stall warning.

KEY TAKEAWAYS

- Unreliable airspeed: disagreements, impossible combinations, contradictory warnings.
- Memory items: autopilot and autothrust off, flight directors off as required, set memory pitch and thrust.
- Stabilise first, then troubleshoot with QRH tables and other data.

KNOWLEDGE CHECK

Shortly after take-off, the airspeed indications disagree and both overspeed and stall warnings occur. What is the first action?

A. Follow the flight director

B. Apply the unreliable airspeed memory items: autopilot and autothrust off, set memory pitch and thrust

C. Pull up to reduce the overspeed

D. Reduce thrust to idle

Answer: B. Contradictory warnings indicate unreliable air data. Disconnect the automation and fly the memory pitch and thrust.

Training material only. It does not replace the Aircraft Flight Manual, FCOM, QRH, MEL or the Domestic Airlines Operations Manual, which always take precedence.